

# BLACK-BOX TEMPLATE — fill the blanks, never invent a layout

Always three parts. Copy the **exact** template the task gives you and replace only the `<fill>` cells. Use `|` separators and keep the pre-filled columns.

## How to know which sentence is which cell (read this first)

The exam gives you the answer template **half filled**. The pre-filled cells are clues, and the task text has **one sentence per class**. So the job is "cross off", not "invent".

| Sentence in the task                                  | Becomes                                               |
|-------------------------------------------------------|-------------------------------------------------------|
| "Valid devices are X and Y; anything else is invalid" | 3 text classes: X (valid), Y (valid), other (invalid) |
| "On X, 2048-4096 gives A. 4097-16384 gives B."        | 2 valid number classes for X                          |
| "On Y, 2048-8192 gives A. 8193-16384 gives B."        | 2 valid number classes for Y                          |
| "<= 2047 is unsupported"                              | 1 invalid class (low side)                            |
| ">= 16385 is unsupported"                             | 1 invalid class (high side)                           |

Steps:

- Underline every range / value sentence in the task and number them. **That count must equal the number of class rows.**
- Write each class on scrap paper (or in the answer box). Cross out the ones the template already shows.
- The classes left over are the blanks. Type them in the `<fill>` cells, keep the `|` separators.
- Part 2: each row must cover a class you have not used yet. Tick a class when a row covers it. Never break two inputs in one row. If a cell is blank, work it out from the other cells: the expected result and the number tell you the device.
- Part 3: the **heading** names the boundary. Find that limit in the text, then write L-1, L, L+1 for that one device.

Worked mapping from the 2026 exam (Graphics Configurator):

|                                                                                                                |                                                                       |                                                       |
|----------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------|-------------------------------------------------------|
| DC1 = "Mobile" is valid                                                                                        | VC1 = <= 2047                                                         | (given in the template)                               |
| DC2 = "Console" is valid                                                                                       | VC2 = Mobile 2048..4096                                               | (performance) <- blank, from sentence "On Mobile ..." |
| DC3 = anything else is invalid                                                                                 | VC3 = Mobile 4097..16384                                              | (quality) (given)                                     |
|                                                                                                                | VC4 = Console 2048..8192                                              | (performance) <- blank                                |
|                                                                                                                | VC5 = Console 8193..16384                                             | (quality) <- blank                                    |
|                                                                                                                | VC6 = >= 16385                                                        | <- blank, from sentence                               |
| "Hardware Memory Ceiling"                                                                                      |                                                                       |                                                       |
| TC1 Mobile + too low   TC2 Mobile 3000 -> VC2   TC3 Mobile 6000 -> VC3   TC4 -> VC4   TC5 Console 12000 -> VC5 |                                                                       |                                                       |
| TC6 -> VC6   TC7 Handheld -> DC3                                                                               | Part 3 heading "Mobile Minimum Engine Baseline" -> 2047 / 2048 / 2049 |                                                       |

# Step 0 — read the rules and mark every limit (30 s)

|                       |                                                   |                                                |
|-----------------------|---------------------------------------------------|------------------------------------------------|
| parameter A (text):   | valid values = _____ , _____                      | anything else (other word, "", null) = INVALID |
| parameter B (number): | system limits <= _____ invalid   >= _____ invalid |                                                |
|                       | band 1: _____ .. _____ -> result _____            | (only for A = _____ )                          |
|                       | band 2: _____ .. _____ -> result _____            | (only for A = _____ )                          |

## Part 1 — equivalence classes

One class per **valid text value**, one catch-all invalid class. For the number: one class per **band per text value**, plus one invalid class per side. Class count check: valid texts + 1 and bands × texts + 2.

```

### `parameterA`
| Class | Valid/Invalid | Inputs |
|---|---|---|
| AC1 | Valid | `` |
| AC2 | Valid | `` |
| AC3 | Invalid | any other value or empty input, e.g. ``, ``, `` |

### `parameterB`
| Class | Valid/Invalid | Inputs |
|---|---|---|
| BC1 | Invalid | `B <= <low limit - 1>` |
| BC2 | Valid | `A = <value 1>` with ` <= B <= <hi>` |
| BC3 | Valid | `A = <value 1>` with `
```

- "Valid" = *accepted and evaluated*, even if the result is "rejected"/"unsupported"-style. Not "the answer is yes".
- Do **not** pair invalid text with number ranges ("PC with 4097-16384" is not needed — it cost marks in 2026).
- Case matters: Mobile ≠ mobile.

## Part 2 — one representative test per class (minimal set, ONE invalid input at a time)

Pick a **middle** value of each band. Every row breaks at most one parameter.

|           |                |                         |                    |                               |
|-----------|----------------|-------------------------|--------------------|-------------------------------|
| Test Case | parameterA     | parameterB              | Expected Result    |                               |
| ---       | ---            | ---                     | ---                |                               |
| TC1       | `<value 1>`    | `<below low limit>`     | `<invalid result>` | <- covers AC1 + BC1           |
| TC2       | `<value 1>`    | `<middle of band 1>`    | `<result 1>`       | <- AC1 + BC2                  |
| TC3       | `<value 1>`    | `<middle of band 2>`    | `<result 2>`       | <- AC1 + BC3                  |
| TC4       | `<value 2>`    | `<middle of band 1>`    | `<result 1>`       | <- AC2 + BC4                  |
| TC5       | `<value 2>`    | `<middle of band 2>`    | `<result 2>`       | <- AC2 + BC5                  |
| TC6       | `<value 2>`    | `<above max>`           | `<invalid result>` | <- AC2 + BC6                  |
| TC7       | `<wrong text>` | `<a perfectly valid B>` | `<invalid result>` | <- AC3 (B must be VALID here) |

Write the **coverage note** if the task allows free text: TC1–TC3 cover AC1 + BC1/2/3 · ... · all N classes, M cases.

## Part 3 — three-point boundary (L−1, L, L+1)

---

1. Read the **heading of Part 3** — it names ONE boundary ("Mobile Minimum Engine Baseline").
2. Find that limit in the text. L = the **first valid value** (limit is "<= 2047 invalid" → L = 2048).
3. Write L−1, L, L+1 for **that device/text value only**, with the result each really gives.

| Test Case | parameterB | Expected Result    |
|-----------|------------|--------------------|
| TC10      | `L-1`      | `<invalid result>` |
| TC11      | `L`        | `<valid result>`   |
| TC12      | `L+1`      | `<valid result>`   |

<- the row that earns the mark (catches >= typed as >)

Upper-limit variant: L = the **last valid value**, so L−1 valid, L valid, L+1 **invalid**. If 1 is the smallest legal value, test 0 / 1 / 2.

## Last check (20 s)

---

- Every expected result is spelled exactly as in the task (unsupported, not invalid).
- No value is reused across rows of the same class (5 and 7 in one band = one row).
- All three values in Part 3 sit around the **same** limit.
- Both text values have their own boundary rows **if the task asks for each**.

## Quick-table version (when the task only says "write test cases for method X")

---

| Class              | Input     | Expected Output |
|--------------------|-----------|-----------------|
| Negative           | -5        | error           |
| Not a whole number | 3.14      | error           |
| Not a number       | "ABC"     | error           |
| Empty / null       | "" / null | error           |
| Too big            | 100       | error           |
| Special value      | 0         | 1               |
| Normal             | 7         | 5040            |
| Edge below limit   | 11        | works           |
| Edge on limit      | 12        | works           |
| Edge above limit   | 13        | error           |